Hip MRI Dictation Cheat Sheet - Expanded Findings & Impressions

Copy-ready language. Replace bracketed text. Use as a workstation reference, not a substitute for surgical correlation or local report style.

Core Measurements

Measurement	Normal / Threshold
Alpha angle	Normal <55 degrees; cam morphology >55 degrees.
Center-edge angle (CEA)	Normal 25-40 degrees; dysplasia <20 degrees; overcoverage >40 degrees.
Tonnis angle	Normal 0-10 degrees; dysplasia >10 degrees.
Acetabular head index (AHI)	Normal >75%; femoral head extrusion index = 100 - AHI.
Femoral version	Normal 10-20 degrees anteversion; >20 degrees excessive anteversion; <10 degrees decreased/retroversion.
Acetabular version	Normal ~15-20 degrees anteversion; retroversion suggested by crossover sign on AP radiograph.

Ficat and Arlet AVN Staging

Stage	Description
0	Preclinical, pre-radiographic; biopsy-only diagnosis.
I	Normal radiograph; MRI shows marrow edema or early necrosis.
II	Sclerosis/cysts on radiograph; no collapse. MRI: double-line sign, preserved sphericity.
III	Subchondral collapse (crescent sign); femoral head still spherical overall.
IV	Femoral head flattening/collapse with secondary osteoarthritis.

Modified Steinberg (University of Pennsylvania) AVN Classification

Stage	Description
0	Normal or nondiagnostic.
I	Normal radiograph; abnormal MRI/bone scan. A <15%, B 15-30%, C >30% head involvement.
II	Lucency and sclerosis on radiograph; no collapse. A/B/C substages by extent.
III	Crescent sign without flattening.
IV	Flattening of femoral head. A <15%, B 15-30%, C >30% surface collapse and <2 mm, 2-4 mm, >4 mm depression.
V	Joint narrowing with or without acetabular changes.
VI	Advanced degenerative changes.

How to Find It — Landmark-Based Identification

This section teaches a systematic, landmark-driven search pattern for hip/pelvis MRI: start with the bony scaffold on coronal images, then use the femoral neck axis to set up oblique and radial planes for the labrum and head-neck junction, and finally interrogate the tendon footprints and neurovascular spaces. Throughout, the convention used here is a **right-hip clock face** where **3 o'clock = anterior**, **12 o'clock = superior**, **6 o'clock = inferior**, and **9 o'clock = posterior**.

Planes and Sequences — Setting Up the Search

💡 PEARL **What each plane buys you**

Build a habit of assigning each structure to its best plane before you start scrolling.

- **Coronal** (fluid-sensitive + T1): marrow disease, superolateral labrum (11–1 o'clock), abductor tendon footprints, sourcil/center-edge geometry.
- **Axial / oblique axial**: anterior and posterior labrum, alpha angle and cam morphology, iliopsoas tendon, hamstring and rectus origins, crossover sign on localizers.
- **Sagittal**: anterior–posterior labral extent, cartilage of the dome, rectus femoris reflected head.

🔍 HOW TO FIND IT **Oblique axial along the femoral neck**

Prescribe an **oblique axial** by drawing the slice plane parallel to the long axis of the **femoral neck** on a coronal image. This single maneuver gives you a true head-neck profile for the **alpha angle** and an undistorted look at the anterosuperior head-neck junction where cam bumps live.

💡 PEARL **Radial sequences are the labrum's best friend**

A **radial** acquisition reslices the joint as spokes of a wheel centered on the **femoral head/neck axis**, so every slice cuts the labrum perpendicular to its base. This eliminates the volume-averaging that fakes (or hides) tears on orthogonal planes and lets you assign findings cleanly to clock positions. MR arthrography improves labral and chondral conspicuity but the landmark logic is identical.

Acetabular Labrum and Its Ligaments

📍 LANDMARK **Clock-face mapping of the labrum**

Anchor your clock to fixed bony cues: **3 o'clock = anterior**, **12 o'clock = superior/lateral apex**, **9 o'clock = posterior**, **6 o'clock = inferior** (the notch bridged by the transverse ligament).

- The **normal labrum** is a small, uniformly **low-signal triangle** on T1/PD continuous with the acetabular rim.
- Most degenerative and traumatic tears occur in the **anterosuperior** quadrant (roughly **11–3 o'clock**).

🔍 HOW TO FIND IT **Anterosuperior labrum**

Track the **anterosuperior labrum** on **oblique axial** and **radial** slices; this is the highest-yield zone for tears in cam/pincer impingement. A tear shows linear high signal extending to the labral surface or base, often with an adjacent **paralabral cyst** that points back to the tear.

⚠️ PITFALL **Sublabral sulcus/recess vs tear**

A **normal sublabral sulcus** is a smooth, shallow cleft at the chondrolabral junction (classically described **posteroinferiorly**, but also seen anteriorly) that does NOT undercut the full labral base.

- **Sulcus**: smooth margins, partial depth, no displacement, no paralabral cyst.
- **Tear**: irregular margins, extends to/through the base, may displace the labrum, often a **paralabral cyst**.

📍 LANDMARK **Transverse acetabular ligament (TAL)**

The **TAL** bridges the **acetabular notch at 6 o'clock**, completing the inferior rim. It is a normal low-signal band — do not call the cleft between the TAL and labrum a tear. The TAL is your landmark for the inferior limit of the labrum.

🔍 HOW TO FIND IT **Ligamentum teres**

Find the **ligamentum teres** running from the **fovea capitis** of the femoral head to the **acetabular notch/TAL**, best seen on **coronal** and **axial** fluid-sensitive images as a low-signal band within fat.

- Normal: thin, taut, uniformly low signal.
- Pathologic: thickening, partial/complete discontinuity, or fluid filling the expected course.

Femoroacetabular Impingement (FAI)

📏 MEASURE Alpha angle (cam morphology)

On an **oblique axial** along the femoral neck, fit a best-fit circle to the **femoral head**; draw the **neck axis** through the center of head and neck; measure the angle between the neck axis and the radius to the point where the head contour first exits the circle (the cam bump).

- Abnormal: **> 55°** (values are best assessed at the anterosuperior junction).

🔍 HOW TO FIND IT Cam morphology

Look for the cam bump at the **anterosuperior head-neck junction** — loss of the normal concave "waist" of the neck. Pair it with adjacent **anterosuperior chondrolabral** damage; cam impingement drives shear at the cartilage that delaminates it from bone.

📏 MEASURE Lateral center-edge angle (LCEA)

On a **coronal** image (or coronal-equivalent), draw a vertical line through the **femoral head center** and a second line to the **lateral edge of the acetabular sourcil**; the angle between them is the LCEA.

- Normal: roughly **25–39°**.
- **Dysplasia**: < 20–25° (undercoverage).
- **Pincer/overcoverage**: > 39–40°.

💡 PEARL Pincer signs — crossover, coxa profunda, protrusio

Pincer is acetabular over-coverage; recognize it by:

- **Crossover sign**: anterior wall line crosses the posterior wall line near the lateral rim (focal anterosuperior over-coverage) — assess on a true AP-equivalent localizer.
- **Coxa profunda**: acetabular floor medial to the ilioischial line.
- **Protrusio**: femoral head medial to the ilioischial line.

🔍 HOW TO FIND IT Chondrolabral junction and cartilage delamination

Interrogate the **acetabular cartilage** right where it meets the labrum (the **chondrolabral junction**) on radial/oblique images. **Delamination** is a fluid/signal cleft splitting cartilage from subchondral bone, often with intact overlying surface — easy to miss without arthrographic contrast or careful radial review. In cam FAI, look anterosuperiorly; in pincer, expect a contrecoup posteroinferior chondral lesion.

The "Rotator Cuff of the Hip" — Abductor Footprints

📍 LANDMARK **Greater trochanter facets**

The greater trochanter has named facets that anchor the abductor footprints; learn which tendon goes where:

- **Anterior facet**: **gluteus minimus** insertion.
- **Lateral facet**: **gluteus medius** (main/lateral tendon).
- **Superoposterior facet**: **gluteus medius** (posterior portion) — covered by the lateral tendon.
- **Posterior facet**: bare area covered by the trochanteric bursa.

🔍 HOW TO FIND IT **Gluteus medius and minimus tendons**

On **coronal** fluid-sensitive images, follow each abductor to its facet; use **axial** images to separate the more anterior **minimus** from the more lateral/posterior **medius**. Tendinosis/tears show footprint signal, thickening, or fluid-filled gaps — the hip analog of supraspinatus pathology.

💡 PEARL **"Rotator cuff of the hip"**

Think of the abductors as the hip's rotator cuff: **gluteus medius/minimus = the cuff**, the **greater trochanter facets = the tuberosity footprints**, and **greater trochanteric pain syndrome** is the clinical correlate of cuff tendinopathy plus bursitis.

📍 LANDMARK **Peritrochanteric bursae**

Three bursae explain lateral hip pain:

- **Subgluteus minimus bursa** (deep, near anterior facet).
- **Subgluteus medius bursa** (between medius tendon and trochanter).
- **Subgluteus maximus / greater trochanteric bursa** (superficial, over the posterior facet) — the largest and most commonly inflamed.

⚠️ PITFALL **Don't overcall T2-bright footprint signal**

A little increased signal at the abductor footprint can be magic-angle effect or mild degeneration. Require a **fluid-signal gap, tendon retraction, or footprint bone irregularity** before calling a tear, and correlate with peritrochanteric edema.

Hamstrings, Rectus Femoris, and Iliopsoas

📍 LANDMARK **Hamstring origin at the ischial tuberosity**

At the **ischial tuberosity**, the hamstrings arise in a reproducible pattern:

- **Conjoint tendon** of **semitendinosus + long head of biceps femoris** arises from the **inferomedial facet**.
- **Semimembranosus** arises more **laterally/anteriorly** from a separate facet.

🔍 HOW TO FIND IT **Hamstring origin**

Use **axial** images at the level of the ischial tuberosity to separate the conjoint tendon from the semimembranosus, then **coronal/sagittal** to measure retraction. Report avulsions by **which tendons** are involved and the **retraction distance**, since this drives surgical decision-making.

📍 LANDMARK **Rectus femoris origin**

The rectus femoris has two heads at the front of the hip:

- **Direct head**: from the **anterior inferior iliac spine (AIIS)**.
- **Indirect (reflected) head**: from the **superior acetabular rim/ridge**, coursing posterolaterally — best traced on **sagittal** and **axial** images.

🔍 HOW TO FIND IT **Iliopsoas tendon and bursa**

Follow the **iliopsoas tendon** anterior to the joint on **axial** images as it passes over the **iliopectineal eminence** toward the **lesser trochanter** insertion. The **iliopsoas bursa** lies deep to the tendon and communicates with the joint in ~15% of people.

- A distended iliopsoas bursa can mimic a cystic mass anterior to the hip — trace it back to the tendon to identify it.

Bone Marrow — AVN, Insufficiency Fracture, Edema Syndromes

🔍 HOW TO FIND IT **Femoral head AVN**

Scan the **anterosuperior femoral head** (the weight-bearing segment) on **coronal** and **sagittal** fluid-sensitive and T1 images. AVN produces a **serpiginous demarcating line** outlining the necrotic segment.

💡 PEARL **Double-line and crescent signs**

Two classic AVN signs:

- **Double-line sign** (T2): an outer **low-signal** sclerotic rim with an inner **high-signal** hyperemic/granulation line — fairly specific for AVN.
- **Crescent sign**: a subchondral fracture line/fluid cleft signaling impending articular collapse.

📏 MEASURE **Quantifying AVN involvement**

Estimate the **percentage of the weight-bearing surface/femoral head** involved and the **subchondral arc/extent**; larger lesions involving the anterolateral weight-bearing column carry higher collapse risk and guide whether joint-preserving surgery is viable.

⚠️ PITFALL **Subchondral insufficiency fracture vs transient edema**

Both show diffuse marrow edema, but they are different entities:

- **Subchondral insufficiency fracture (SIF)**: a **low-signal subchondral line paralleling the articular surface** within the edema, often with surrounding edema and risk of collapse — look hard for the fracture line.
- **Transient osteoporosis / bone marrow edema syndrome**: **diffuse edema of head ± neck WITHOUT a subchondral fracture line**, typically self-limited.
- The presence or absence of the **subchondral line** is the discriminator — and missing an SIF risks progression to collapse.

SI Joints, Pubic Symphysis, and the Deep Gluteal Space

🔍 HOW TO FIND IT **Sacroiliac joints**

Evaluate the **SI joints** on oblique coronal/axial fluid-sensitive and T1 images for **periarticular marrow edema (osteitis)**, erosions, and subchondral sclerosis. Symmetric vs asymmetric involvement plus marrow edema helps separate inflammatory sacroiliitis from degenerative or stress-related change.

📍 LANDMARK **Pubic symphysis and athletic pubalgia**

Center on the **pubic symphysis** and the **rectus abdominis–adductor longus aponeurosis** that blends across its anterior margin.

- **Athletic pubalgia / "sports hernia"**: disruption or signal at the **rectus abdominis/adductor aponeurosis**, the **secondary cleft sign**, and **parasymphyseal marrow edema** — there is no true fascial hernia, so trace the aponeurotic plate carefully.

🔍 HOW TO FIND IT **Deep gluteal space and sciatic nerve**

Locate the **deep gluteal space** posterior to the hip, bounded deep by the posterior acetabular column/capsule and superficial by gluteus maximus. Trace the **sciatic nerve** as it passes the **greater sciatic notch** and the **piriformis** muscle on **axial** images.

- Look for nerve enlargement, signal change, or perineural scarring/edema where the nerve crosses piriformis — the anatomic substrate of **deep gluteal syndrome ("piriformis syndrome")**.

Modified Outerbridge Chondromalacia Grading - Copy-Ready

Use this for any articular surface. Replace [surface/compartment] with the specific location: patellar median ridge, medial femoral condyle weightbearing surface, anterosuperior acetabulum, radiocapitellar joint, tibiotalar dome, first MTP joint, etc.

Grade	MRI shorthand
0	Normal cartilage.
1	Signal heterogeneity/softening or swelling; surface intact.
2	Superficial fissuring/fraying/partial-thickness defect <50% cartilage depth.
3	Deep fissuring/partial-thickness defect >50% cartilage depth, not exposing bone.
4	Full-thickness cartilage loss with exposed subchondral bone, often edema/cysts.

Grade 0 - normal cartilage

F: Articular cartilage of the [surface/compartment] is preserved in thickness and signal without focal fissuring, delamination, or subchondral marrow abnormality.

I: No chondral defect of the [surface/compartment].

Grade 1 - chondral signal change / softening

F: Mild focal cartilage signal heterogeneity/softening of the [surface/compartment] with preserved cartilage thickness and intact articular surface. No subchondral marrow edema.

I: Grade 1 chondromalacia of the [surface/compartment].

Grade 2 - superficial partial-thickness chondral loss

F: Superficial chondral fissuring/fraying of the [surface/compartment] involving less than 50% of cartilage thickness. No full-thickness defect or subchondral marrow edema.

I: Grade 2 chondromalacia of the [surface/compartment].

Grade 3 - deep partial-thickness chondral loss

F: Deep partial-thickness chondral fissuring/defect of the [surface/compartment] involving greater than 50% of cartilage thickness without exposed subchondral bone. [Mild adjacent subchondral marrow edema.]

I: Grade 3 chondromalacia of the [surface/compartment].

Grade 4 - full-thickness chondral loss

F: Full-thickness cartilage loss/fissuring of the [surface/compartment] measuring [X] x [Y] mm with exposed subchondral bone and [subchondral marrow edema/cystic change]. [Small unstable chondral flap/loose body if present.]

I: Grade 4 chondromalacia/full-thickness chondral defect of the [surface/compartment].

Hip MRI Tells

- Labral tear + cam + chondral delamination = classic FAI triad.
- Wave sign = delaminated acetabular cartilage lifting off subchondral bone with fluid undermining the flap.
- Paralabral cyst: always search for the associated labral tear at the adjacent clock position.
- Herniation pit: fibrocystic change at the anterosuperior femoral head-neck junction; marker for cam morphology and chronic abutment.
- Double-line sign on T2 = pathognomonic for AVN (outer low-signal sclerosis, inner high-signal granulation tissue).
- Crescent sign = subchondral fracture/collapse; thin linear T1 hypointensity paralleling the articular surface with fluid signal deep to it.
- QF edema: check ischiofemoral space measurement before calling isolated muscle strain.
- Athletic pubalgia: secondary cleft sign at the rectus abdominis/adductor longus aponeurosis insertion on the pubic symphysis.
- Sublabral sulcus at the posteroinferior labrum (5-7 o'clock) = normal variant, not a tear.
- Absent perilabral recess anteroinferiorly is normal; do not overcall.

Normal / Negative Templates

Normal hip

> **F:** Hip joint alignment is anatomic. Femoral head sphericity is preserved without subchondral signal abnormality. Acetabular and femoral head articular cartilage is preserved in thickness and signal. No joint effusion or synovitis. Periarticular tendons and musculature demonstrate normal signal and morphology. No fracture, marrow replacement, or aggressive osseous lesion.

> **I:** Normal MRI of the hip.

Normal labrum on nonarthrogram

> **F:** No displaced labral tear, paralabral cyst, or frank labral detachment identified on this nonarthrographic examination. Note: nondisplaced labral tears and intrasubstance degeneration are better evaluated with direct MR arthrography.

> **I:** No discrete labral tear identified by nonarthrographic MRI. MR arthrogram may be considered if clinical suspicion for labral pathology persists.

Useful Caveats

- **Sublabral sulcus:** A smooth recess at the posteroinferior labrum (approximately 5-7 o'clock) is a normal variant and should not be called a tear. True tears have irregular margins, paralabral cysts, or adjacent chondral injury.
- **Labral degeneration vs. tear on non-arthrogram:** Intermediate intrasubstance labral signal without a fluid-signal cleft reaching the labral surface favors degeneration over discrete tear. Direct MR arthrography is more sensitive for nondisplaced tears.

- **Transient osteoporosis vs. AVN:** Transient osteoporosis shows diffuse marrow edema pattern in the femoral head/neck without a focal subchondral lesion or double-line sign; it is self-limiting. AVN shows focal subchondral involvement, serpiginous demarcation, and the double-line sign.
- **Femoral neck stress fracture location matters:** Compression-side (medial/inferior cortex) injuries are generally managed conservatively; tension-side (lateral/superior cortex) injuries are at higher risk for completion and may require fixation.
- **Non-arthrogram labral sensitivity:** MRI without intra-articular contrast has lower sensitivity for small nondisplaced labral tears. State this limitation when findings are equivocal.

Labrum / FAI / Cartilage

Intact labrum

> **F:** The acetabular labrum is triangular in morphology with uniform low T1 and T2 signal throughout. No surface irregularity, detachment, paralabral cyst, or adjacent chondral injury. The labral-chondral junction is smooth.

> **I:** Intact acetabular labrum without tear or degeneration.

Labral degeneration (intrasubstance signal)

> **F:** Intermediate intrasubstance signal on proton density/T2-weighted sequences within the [anterosuperior/superior] acetabular labrum without a discrete fluid-signal cleft extending to the labral surface. Labral morphology is mildly blunted/irregular. No paralabral cyst.

> **I:** Intrasubstance labral degeneration of the [anterosuperior/superior] labrum. No discrete tear by nonarthrographic criteria.

Anterosuperior labral tear with paralabral cyst

> **F:** Irregular fluid-signal tear of the anterosuperior acetabular labrum at [11 to 1 o'clock] with a [X] mm paralabral cyst extending [superiorly/laterally]. Adjacent acetabular cartilage is [intact/shows early fissuring].

> **I:** Anterosuperior acetabular labral tear with paralabral cyst at [11-1 o'clock].

Superior labral tear

> **F:** Fluid-signal cleft undermining the superior acetabular labrum at [12 to 2 o'clock] with labral detachment from the acetabular rim. [No/small] paralabral cyst. Adjacent superior acetabular cartilage demonstrates [no injury/early fissuring/delamination].

> **I:** Superior acetabular labral tear at [12-2 o'clock].

Posteroinferior labral tear

> **F:** Irregular fluid-signal cleft at the posteroinferior acetabular labrum at [7-9 o'clock] with adjacent periosteal edema and mild subchondral marrow signal change. No paralabral cyst.

> **I:** Posteroinferior acetabular labral tear with periosteal edema. [Correlate for posterior impingement or instability pattern.]

Detached labral fragment with wave sign and chondral delamination

> **F:** Detached labral fragment at the [anterosuperior] acetabulum with displaced labral tissue and undulating/delaminated acetabular cartilage (wave sign) measuring [X] mm in extent. Fluid undermines the acetabular cartilage with separation from subchondral bone.

> **I:** Anterosuperior labral tear with detached fragment, chondrolabral delamination, and wave sign.

Cam FAI

> **F:** Asphericity and osseous prominence at the anterolateral femoral head-neck junction with elevated alpha angle of [X] degrees (measured on oblique axial). [Small fibrocystic change/synovial herniation pit at the anterosuperior head-neck junction measuring X mm.] Associated anterosuperior labral tear and adjacent acetabular chondral [delamination/fissuring].

> **I:** Cam-type femoroacetabular impingement with elevated alpha angle, anterosuperior labral tear, and chondral injury. [Synovial herniation pit.]

Pincer FAI

> **F:** Acetabular overcoverage with center-edge angle of [X] degrees (>40 degrees), prominent anterior acetabular rim, and crossover sign on coronal images. Labral degeneration/tear at [12-3 o'clock]. [Posteroinferior contrecoup chondral wear/marrow edema.]

> **I:** Pincer-type femoroacetabular impingement pattern with acetabular overcoverage and labral pathology.

Combined FAI

> **F:** Combined cam and pincer morphology. Aspherical femoral head-neck junction with alpha angle of [X] degrees and acetabular overcoverage with center-edge angle of [X] degrees. Anterosuperior labral tear with adjacent chondral delamination. [Posteroinferior contrecoup cartilage wear.]

> **I:** Combined cam and pincer femoroacetabular impingement with labral tear and chondral injury.

Mild femoral head-neck flattening without bump

> **F:** Mild asphericity/flattening of the anterolateral femoral head-neck junction without a discrete osseous bump. Alpha angle measures [X] degrees (borderline/mildly elevated). No labral tear or chondral injury identified.

> **I:** Mild cam-type morphology without definite impingement-related labral or chondral damage. Clinical correlation recommended.

Focal acetabular cartilage thinning

> **F:** Focal cartilage thinning/fissuring at the [anterosuperior/weightbearing] acetabulum measuring [X] mm in extent without full-thickness loss. Mild adjacent subchondral marrow edema. Labrum is [intact/torn].

> **I:** Focal partial-thickness acetabular chondral loss at the [anterosuperior] acetabulum. [Grade 2/3 chondromalacia.]

Chondrolabral delamination with wave sign

> **F:** Chondrolabral separation at the anterosuperior acetabulum with fluid undermining a delaminating cartilage flap measuring [X] mm (wave sign). The detached cartilage is lifted from the subchondral bone plate. Associated labral tear at [clock position].

> **I:** Anterosuperior acetabular chondrolabral delamination with wave sign, associated with labral tear.

Diffuse femoral head cartilage thinning

> **F:** Diffuse thinning of the femoral head articular cartilage, most pronounced at the [weightbearing/anterosuperior] surface. Subchondral marrow edema and early cystic change are noted. Marginal osteophytes.

> **I:** Diffuse femoral head chondral thinning consistent with [moderate/severe] osteoarthrosis.

Full-thickness acetabular cartilage loss

> **F:** Full-thickness cartilage loss at the [anterosuperior/weightbearing] acetabulum measuring [X] x [Y] mm with exposed subchondral bone plate, subchondral marrow edema, and [subchondral cystic change]. [Associated labral tear.]

> **I:** Full-thickness acetabular chondral defect (Outerbridge grade 4) at the [anterosuperior] acetabulum.

Tendons / Bursae

Normal gluteal tendons

> **F:** Gluteus medius and minimus tendons demonstrate normal signal and morphology at their greater trochanter insertions without tendinosis, tear, or peritendinous edema. Greater trochanteric bursae are not distended.

> **I:** Normal gluteus medius and minimus tendons. No trochanteric bursitis.

Gluteus medius tendinosis with trochanteric bursitis

> **F:** Gluteus medius tendon is thickened with intermediate intrasubstance signal at the lateral facet of the greater trochanter without discrete fiber discontinuity. Fluid and synovial thickening in the greater trochanteric/subgluteus medius bursa with lateral peritrochanteric soft tissue edema.

> **I:** Gluteus medius tendinosis at the lateral facet with greater trochanteric bursitis.

Gluteus medius partial undersurface tear

> **F:** Focal fluid-signal cleft at the undersurface/deep fibers of the gluteus medius tendon at the lateral facet of the greater trochanter involving approximately [X]% of tendon thickness. Superficial fibers remain intact. Mild peritrochanteric edema/bursal fluid.

> **I:** Partial-thickness undersurface tear of the gluteus medius tendon at the lateral facet.

Gluteus medius full-thickness tear

> **F:** Full-thickness discontinuity of the gluteus medius tendon at the lateral facet of the greater trochanter with [X] cm of lateral/proximal retraction. [Moderate/severe] fatty infiltration and volume loss of the gluteus medius muscle belly. Trochanteric bursal fluid.

> **I:** Full-thickness gluteus medius tendon tear with retraction and fatty atrophy.

Gluteus minimus tendinosis

> **F:** Gluteus minimus tendon is thickened with intermediate signal at the anterior facet of the greater trochanter. No discrete fiber discontinuity. Mild subgluteus minimus bursal fluid.

> **I:** Gluteus minimus tendinosis at the anterior facet.

Trochanteric bursitis

> **F:** Fluid and synovial thickening in the greater trochanteric bursa with adjacent lateral peritrochanteric soft tissue edema. Gluteal tendons are [intact/show tendinosis/torn].

> **I:** Greater trochanteric bursitis/peritrochanteric pain syndrome.

Normal hamstring origin

> **F:** Proximal hamstring tendons (conjoint tendon of the biceps femoris and semitendinosus, and the semimembranosus tendon) at the ischial tuberosity demonstrate normal signal, morphology, and attachment. No peritendinous edema or ischial bursitis.

> **I:** Normal proximal hamstring tendon origin. No tear or tendinosis.

Hamstring tendinosis with ischial bursitis

> **F:** Thickening and intermediate intrasubstance signal of the proximal hamstring tendons at the ischial tuberosity without discrete fiber disruption. Fluid/edema in the ischial bursa deep to the hamstring origin.

> **I:** Proximal hamstring tendinosis at the ischial tuberosity with ischial bursitis.

Partial hamstring tear

> **F:** Partial-thickness fiber disruption of the [conjoint/semimembranosus] tendon at the ischial tuberosity with surrounding T2 hyperintense edema/hemorrhage. Approximately [X]% of the tendon cross-section is involved. Remaining fibers are in continuity.

> **I:** Partial-thickness proximal hamstring tendon tear at the ischial tuberosity with peritendinous edema.

Complete hamstring tear with retraction

> **F:** Complete tear of the conjoint/semitendinosus-biceps femoris and semimembranosus tendons from the ischial tuberosity with [X] cm of distal retraction and surrounding T2 hyperintense hematoma. The sciatic nerve is [normally positioned/displaced by hematoma/encased in scar; note proximity].

> **I:** Complete proximal hamstring avulsion tear with [X] cm retraction and hematoma. [Note sciatic nerve relationship.]

Hamstring avulsion fracture

> **F:** Avulsion fracture fragment from the ischial tuberosity measuring [X] mm displaced [inferolaterally] by [X] cm with attached hamstring tendon origin. Surrounding marrow edema and soft tissue hemorrhage/edema. Sciatic nerve [is/is not] displaced.

> **I:** Ischial tuberosity avulsion fracture at the hamstring origin with [X] cm displacement.

Normal iliopsoas

> **F:** Iliopsoas tendon is intact with normal signal and morphology at the lesser trochanter insertion. Iliopsoas bursa is not distended.

> **I:** Normal iliopsoas tendon. No bursitis.

Iliopsoas tendinosis with bursitis

> **F:** Iliopsoas tendon is thickened with intermediate intrasubstance signal near the lesser trochanter insertion. Distended iliopsoas bursa with T2 hyperintense fluid deep to the tendon measuring [X] mm.

I: Iliopsoas tendinosis with iliopsoas bursitis.

Iliopsoas bursitis alone

F: Distended iliopsoas bursa with T2 hyperintense fluid measuring [X] cm anterior to the hip joint capsule. Iliopsoas tendon is intact without tendinosis.

I: Iliopsoas bursitis without tendon tear.

Ischiofemoral Impingement

Normal ischiofemoral space

F: Ischiofemoral space measures [X] mm (normal >15 mm) and quadratus femoris space measures [X] mm (normal >10 mm). Quadratus femoris muscle demonstrates normal signal without edema or fatty replacement.

I: Normal ischiofemoral space. No impingement.

Ischiofemoral impingement, acute

F: Narrowed ischiofemoral space measuring [X] mm (normal >15 mm) and quadratus femoris space measuring [X] mm (normal >10 mm). Quadratus femoris muscle demonstrates diffuse T2 hyperintense edema without fatty replacement, compatible with acute/subacute impingement.

I: Ischiofemoral impingement with quadratus femoris edema.

Ischiofemoral impingement, chronic

F: Narrowed ischiofemoral space measuring [X] mm. Quadratus femoris muscle demonstrates T1 hyperintense fatty replacement and volume loss without significant edema, compatible with chronic impingement and denervation-type change.

I: Chronic ischiofemoral impingement with fatty replacement of the quadratus femoris.

Nerves

Normal sciatic nerve and piriformis

F: Sciatic nerve demonstrates normal caliber and signal in the greater sciatic notch and along the posterior hip. Piriformis muscle is symmetric in size and signal.

I: Normal sciatic nerve. Normal piriformis muscle.

Piriformis asymmetry with sciatic nerve thickening

F: Piriformis muscle is asymmetrically [enlarged/edematous] compared to the contralateral side. Sciatic nerve is mildly thickened/hyperintense on fluid-sensitive sequences at the level of the piriformis. No discrete mass lesion.

I: Piriformis asymmetry with sciatic nerve thickening, compatible with piriformis syndrome. Clinical correlation recommended.

Sciatic neuritis

F: Sciatic nerve demonstrates increased T2 signal and mild fascicular thickening at the level of the [piriformis/ischial tuberosity/posterior hip] with perineural edema in the surrounding fat. No discrete compressive mass.

I: Sciatic neuritis with perineural edema. [Correlate with clinical radiculopathy/neuropathy.]

Sciatic nerve through piriformis (variant)

F: Anatomic variant with the [peroneal division of the] sciatic nerve coursing through the piriformis muscle belly rather than deep to it. The nerve demonstrates [normal signal/mild T2 hyperintensity].

I: Variant sciatic nerve course through the piriformis muscle. [Possible contributing factor if clinically symptomatic.]

Bone / AVN / Stress

Normal femoral head

F: Femoral head is spherical with preserved contour and uniform marrow signal on T1 and T2 sequences. No subchondral fracture, marrow edema, or serpiginous demarcation. No collapse.

I: Normal femoral head without avascular necrosis or fracture.

Early AVN - double-line sign, preserved sphericity

F: Serpiginous/geographic low-signal line on T1 in the [anterosuperior/weightbearing] femoral head with characteristic double-line sign on T2 (outer low-signal rim, inner high-signal rim) involving approximately [X]% of the weightbearing surface. Femoral head sphericity is preserved without subchondral fracture or collapse.

I: Avascular necrosis of the femoral head without collapse (Ficat stage II / Steinberg stage II). Preserved femoral head contour.

AVN without collapse, with sclerosis and cysts

F: Geographic area of marrow signal abnormality in the [anterosuperior] femoral head with serpiginous low-signal demarcation on T1, double-line sign on T2, involving approximately [X]% of the weightbearing surface. Sclerotic reactive margin and small subchondral cysts are present. No crescent sign. Femoral head contour is preserved.

I: Avascular necrosis of the femoral head without collapse, with reactive sclerosis and subchondral cysts (Ficat stage II).

AVN with crescent sign - early collapse

F: Subchondral fracture line (crescent sign) paralleling the articular surface of the [anterosuperior] femoral head as a thin linear T1 hypointensity with fluid signal deep to it. Underlying serpiginous demarcation of avascular necrosis involves approximately [X]% of the weightbearing surface. Mild flattening/depression of the articular surface measuring [X] mm.

I: Avascular necrosis with crescent sign and early subchondral collapse (Ficat stage III).

AVN with complete collapse and secondary OA

> **F:** Complete collapse and flattening of the femoral head with loss of normal sphericity. Diffuse low T1 marrow signal with cystic change and sclerosis. Secondary acetabular cartilage loss, marginal osteophytes, and subchondral cystic change. Joint effusion with synovitis.

> **I:** Avascular necrosis with femoral head collapse and secondary osteoarthritis (Ficat stage IV).

Subchondral fracture without double-line sign

> **F:** Subchondral linear low-signal fracture line paralleling the articular surface of the [anterosuperior/weightbearing] femoral head with surrounding marrow edema on fluid-sensitive sequences. No serpiginous demarcation, double-line sign, or geographic marrow replacement to suggest AVN. Femoral head contour is preserved.

> **I:** Subchondral insufficiency fracture of the femoral head. No findings to suggest avascular necrosis. [Differential includes early AVN; clinical and imaging follow-up recommended.]

Compression-side femoral neck stress fracture

> **F:** Linear low-signal fracture line oriented perpendicular to the compression (medial/inferior) cortex of the femoral neck with surrounding T2 hyperintense marrow edema and periosteal edema. Fracture involves approximately [X]% of the femoral neck width. No fracture at the tension (lateral/superior) cortex.

> **I:** Compression-side femoral neck stress fracture. [Low risk for completion; typically managed conservatively.]

Tension-side femoral neck stress fracture

> **F:** Linear low-signal fracture line oriented perpendicular to the tension (lateral/superior) cortex of the femoral neck with surrounding marrow edema and periosteal reaction. Fracture involves approximately [X]% of the femoral neck width. [No/mild] displacement.

> **I:** Tension-side femoral neck stress fracture. [Higher risk for fracture completion; surgical consultation recommended.]

Femoral neck stress reaction without fracture line

> **F:** Ill-defined T2 hyperintense/T1 hypointense marrow edema within the femoral neck without a discrete fracture line. Mild periosteal edema along the [medial/lateral] cortex. No cortical disruption.

> **I:** Femoral neck stress reaction without visible fracture line. [Recommend activity modification and follow-up imaging if symptoms persist.]

Osteoarthritis

> **F:** Diffuse hip chondral thinning/loss with marginal osteophytes, subchondral cystic change/marrow edema, degenerative labral tearing, and small effusion/synovitis. [Femoral head migration is [superolateral/medial].]

> **I:** [Mild/moderate/severe] hip osteoarthrosis with degenerative labral tearing.

Athletic Pubalgia / Groin

Athletic pubalgia - rectus/adductor aponeurosis

> **F:** T2 hyperintense signal and thickening at the rectus abdominis/adductor longus common aponeurotic insertion on the anterior pubic symphysis. Linear fluid-signal secondary cleft extends along the aponeurosis. Mild parasymphyseal marrow edema. Adductor longus tendon origin demonstrates [tendinosis/partial fiber disruption].

I: Athletic pubalgia with injury at the rectus abdominis/adductor longus aponeurosis and secondary cleft sign. [Associated adductor tendinopathy.]

Muscle strain - hip/groin

F: T2 hyperintense intramuscular edema and [mild/moderate] fiber disruption within the [rectus femoris/adductor longus/iliopsoas/sartorius] muscle at the [myotendinous junction/muscle belly] without complete discontinuity. [Small intramuscular hematoma measuring X cm.]

I: [Grade 1/2] strain of the [muscle] at the [myotendinous junction].

Effusion / Other

Trace effusion

F: Trace physiologic fluid in the hip joint. No synovial thickening or capsular distention.

I: Trace hip joint effusion, likely physiologic.

Small effusion with synovial thickening

F: Small hip joint effusion with mild synovial thickening. Anterior capsule is not distended.

I: Small hip joint effusion with mild synovitis.

Moderate effusion

F: Moderate hip joint effusion distending the anterior joint capsule/anterior recess. [Synovial thickening is present.] [No intra-articular loose body.]

I: Moderate hip joint effusion [with synovitis]. [Differential includes inflammatory, infectious, or degenerative etiology; correlate clinically.]

Search Pattern

- **1. Check history, indication, and priors.**
- **2. Assess study adequacy, technique, and limitations.**
- **3. Look at localizers** for incidental pathology.
- **4. Assess bone cortex and marrow:**
- Cortical breaks/fractures, osteophytes, loose bodies, heterotopic ossification; marrow replacement; hematopoietic marrow.
- Lower lumbar spine (disc degeneration, stenosis); sacrum (insufficiency fractures); bony pelvis.
- Acetabulum (developmental dysplasia = shallow acetabulum, superior aspect should be horizontal on coronals; pincer FAI = focal retroversion; osteophytes).
- Femurs (SCFE or AVN in appropriate populations; osteochondral lesions; femoral head deformity/collapse; CAM-type FAI at head-neck junction; neck and shaft fracture/deformity).
- **5. Assess tendinous insertions:**
- Iliopectineal line/pubic symphysis (rectus abdominis, adductors — sports hernias).

- ASIS (sartorius); AIIS (rectus femoris indirect head); superior acetabular labrum (rectus femoris direct head).
- Ischial tuberosity (hamstrings); iliac crests (quadratus lumborum).
- Greater trochanter (gluteus medius/minimus, external rotators); lesser trochanter (iliopsoas).
- Look for fluid along iliopsoas course and at greater trochanter (bursitis).
- **6. Assess sacroiliac joints:** inflammatory change bilaterally (axials + coronals); diastasis, erosion/destruction, ankylosis, degenerative changes.
- **7. Assess pubic symphysis:** erosions, degenerative change, osteitis, diastasis.
- **8. Assess femoroacetabular joints:**
- Effusion; labrum (most commonly anterosuperior quadrant).
- Cartilage (best on arthrographic images if obtained, otherwise PD and fluid-sensitive sagittals and coronals).
- Subchondral marrow changes.
- **9. In the pelvis, don't forget visceral structures:** bowel (mass, inflammation, fistula); reproductive organs; ureters, bladder, urethra; retroperitoneum and pelvic sidewalls (adenopathy); ascites/collections/mass lesions.
- **10. Assess subcutaneous tissues and musculature:** T1 precontrast, fluid-sensitive; muscle signal; collections/effusions.
- **11. Double check** marrow signal, muscles, and subcutaneous tissues.
- **12. Proofread report.**

Infection Search Pattern

- **1. Marrow/bone changes:**
- Marrow replacement (darker than muscle/disc on T1; geographic pattern more predictive than reticular/periarticular).
- Marrow edema (isolated finding raises concern in diabetics/high-risk).
- Bone erosion (loss of dark T1/T2 cortical line).
- **2. Surrounding inflammatory changes:** fascia, musculature, subcutaneous fat edema/enhancement.
- **3. Susceptibility suggesting air:** in collections, soft tissues, along fascial planes, in bone.
- **4. Abnormal enhancement:**
- Rim-enhancing collection = abscess.
- Superficial lesion without enhancement = ulcer.
- Enhancement along fascial planes = infectious/inflammatory fasciitis.
- Non-enhancement of normal structures = ischemia.
- Other patterns of enhancement may not be specific.
- **5. Differentiating osteomyelitis from neuropathic joint:**
- Joint effusion with marrow replacement on both sides of joint.
- Fluid collections/abscesses (periprosthetic, subperiosteal, intraosseous/Brodie's abscess).
- Sinus tract connecting abnormal marrow signal to skin surface.
- Disappearance of subchondral cysts on sequential imaging.

- Progressive marrow/soft tissue changes greater than expected on sequential imaging.

Source Notes

Built as a practical dictation aid from standard MSK MRI reporting patterns, review-level radiology references (including Steinberg/Ficat AVN staging, ischiofemoral impingement criteria, athletic pubalgia imaging findings), and the existing shoulder cheat-sheet style. PowerScribe build-script template choices integrated as F/I pairs. Use surgical correlation and local report conventions when needed.